Human-AI Agent Collaboration

Collaborating with AI Agents

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AI-generated headline

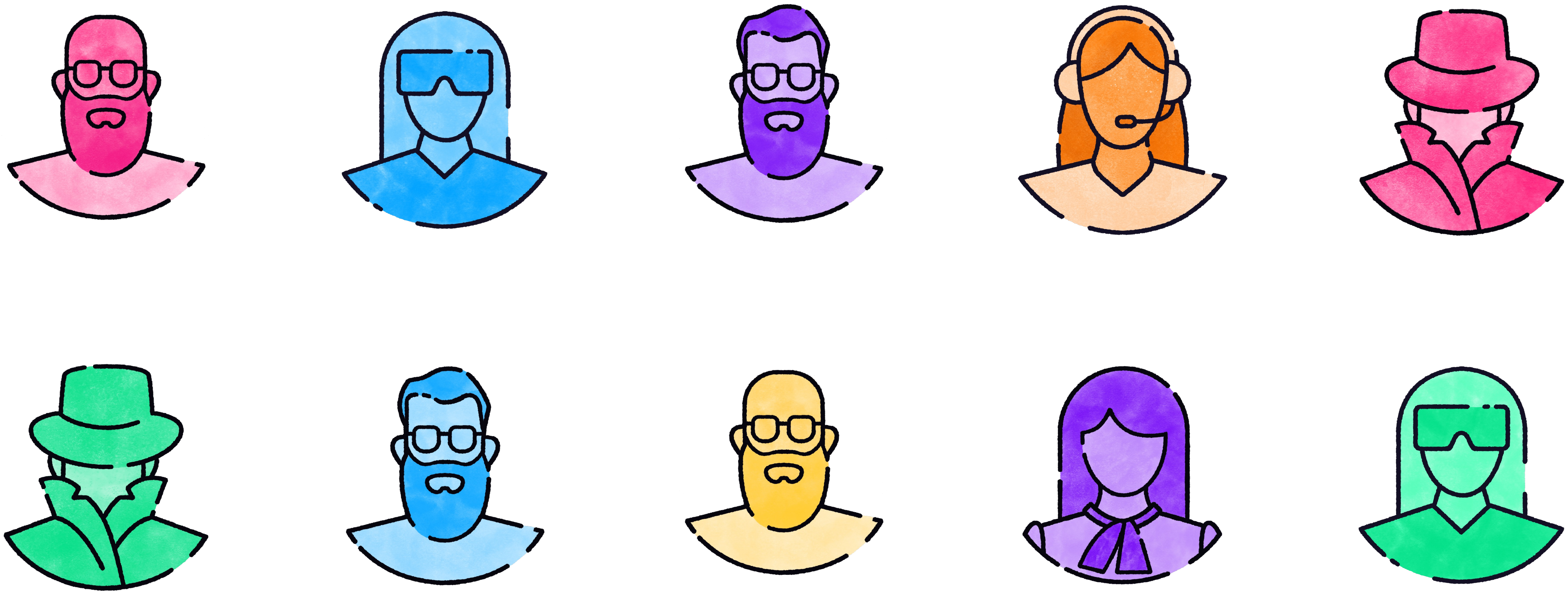
“AI Code Generation: Software Engineers are no longer needed”

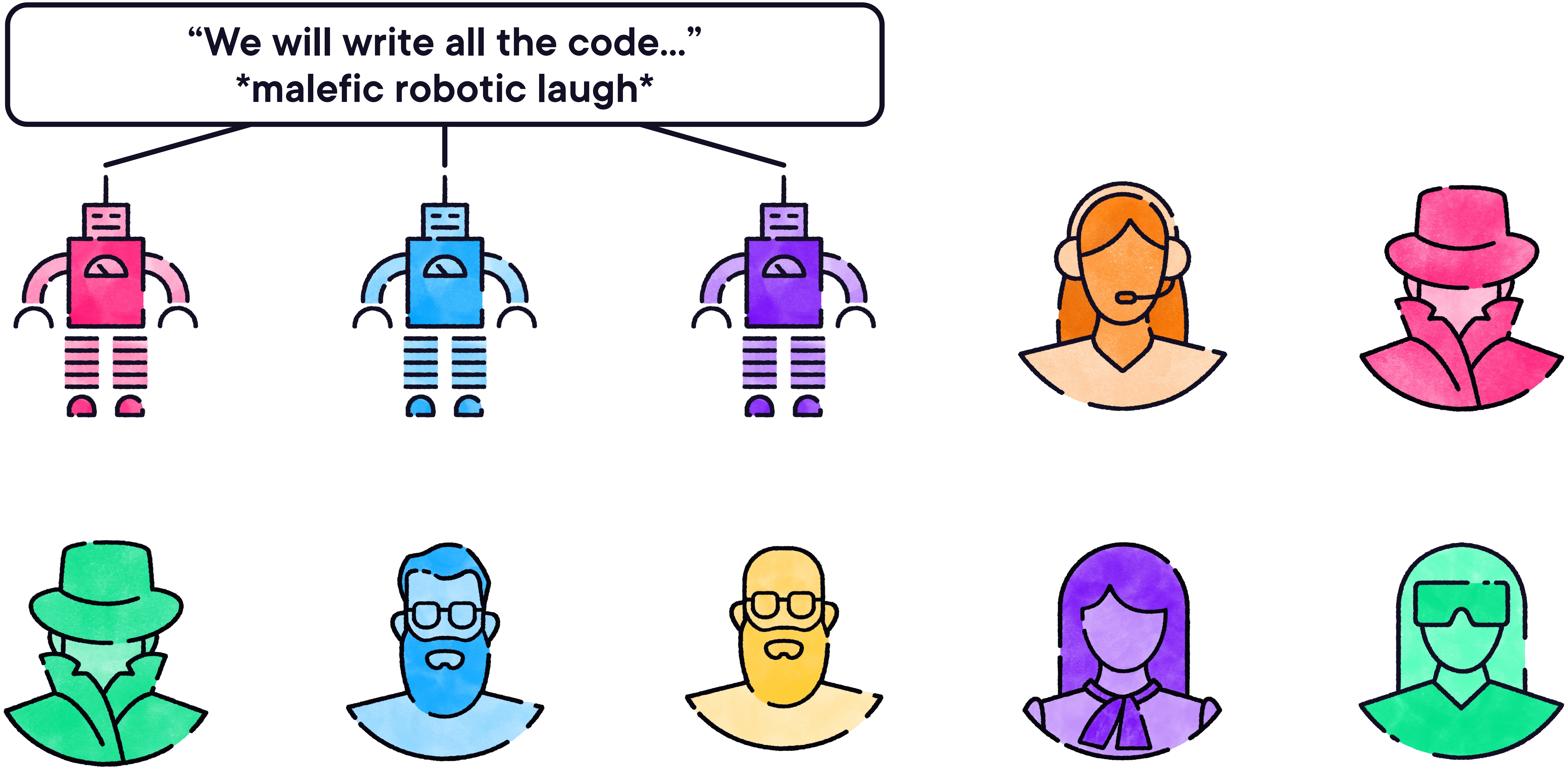
Mark Zuckerberg – CEO of Meta

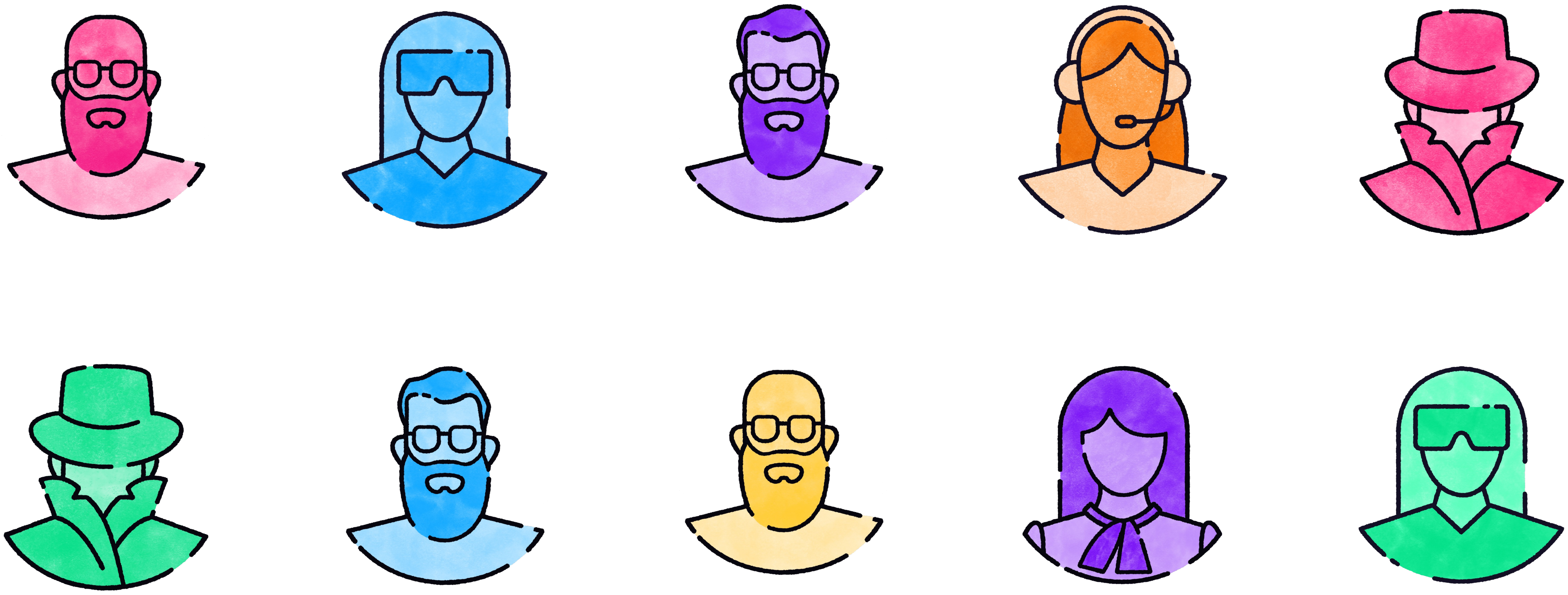
“We at Meta … are going to have an AI that can effectively be a sort of mid-level engineer”

Satya Nadella – CEO of Microsoft

“I’d say maybe 20%, 30% of the code that is inside of our repos today and some of our projects are probably all written by software.”

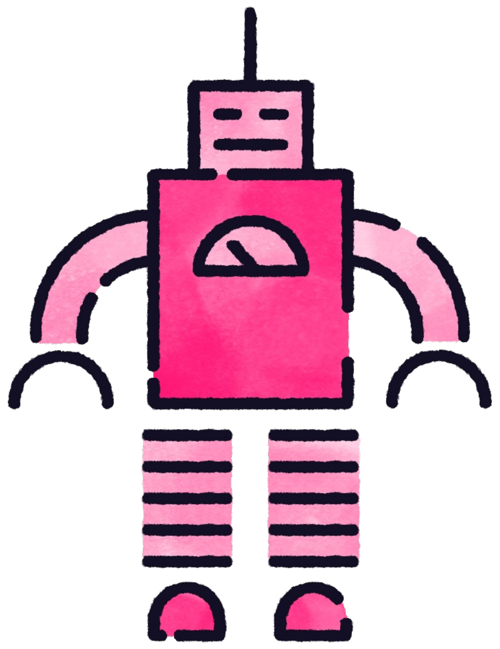



"We will write all the code..."
malefic robotic laugh



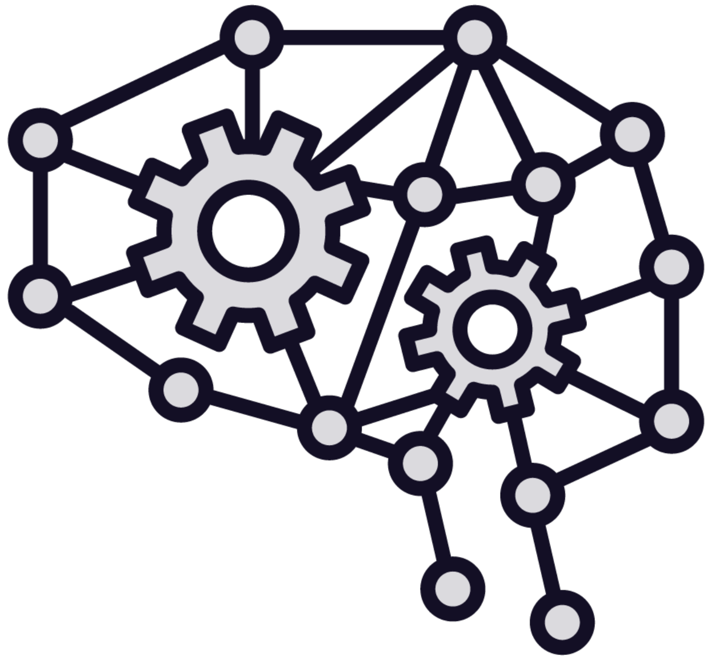
"Can you help me write a new function?"

"Absolutely, tell me more about it …"



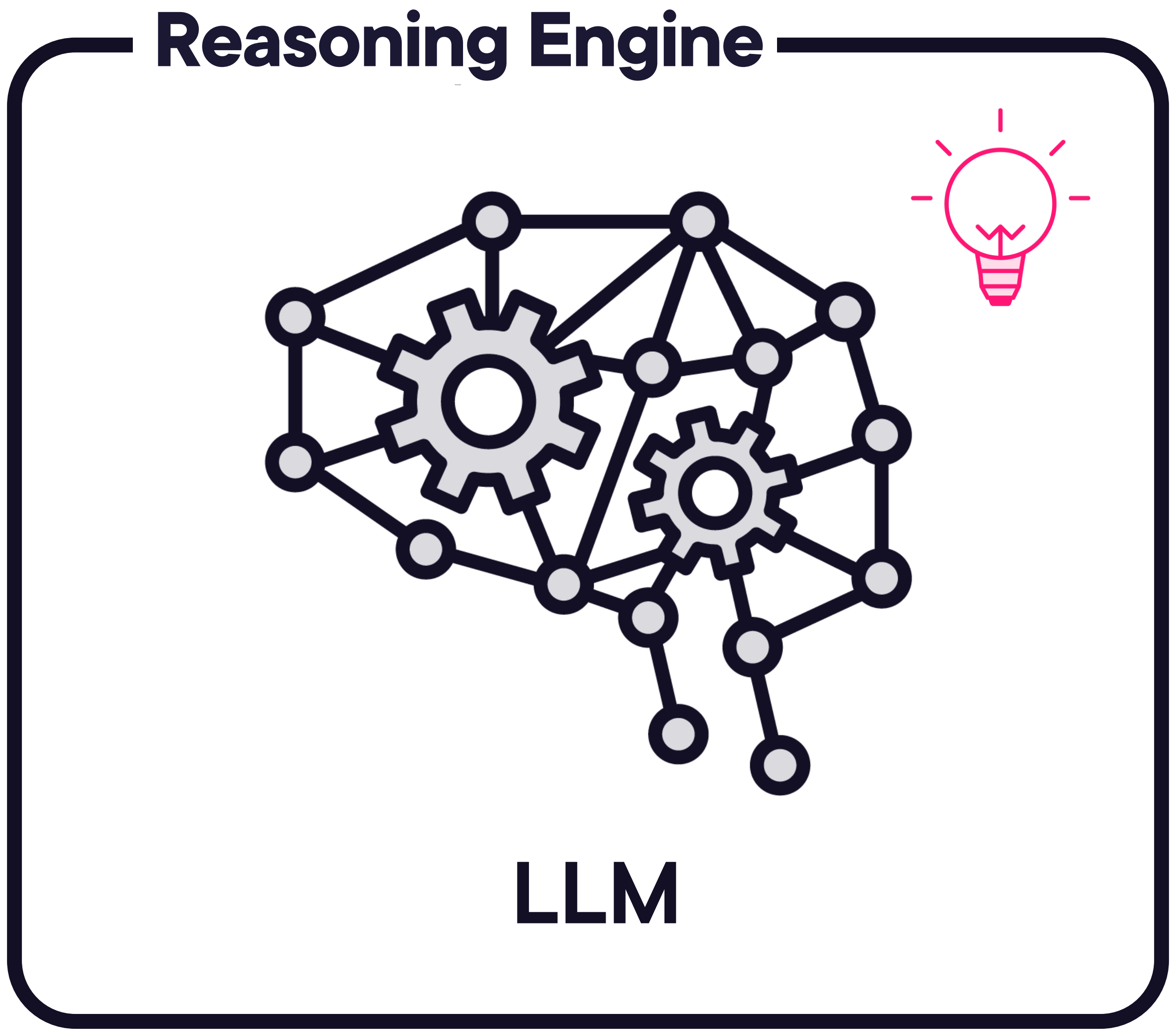
AI Agent

AI Agents



LLM

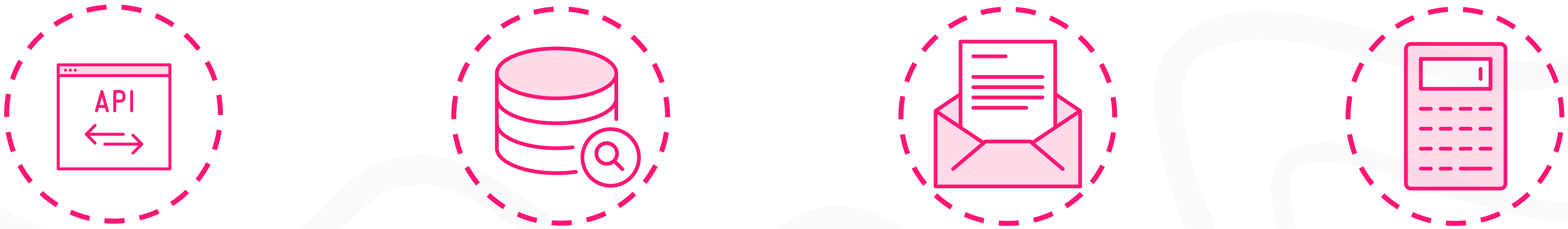
AI Agents



AI Agents

LLM

Tool executor



AI Agents

LLM

Tool executor

Code executor

Human-in-the-loop

Type of AI Agents

General purpose

OpenAI ChatGPT

Anthropic Claude

Google Gemini

Type of AI Agents

General purpose

Domain specific

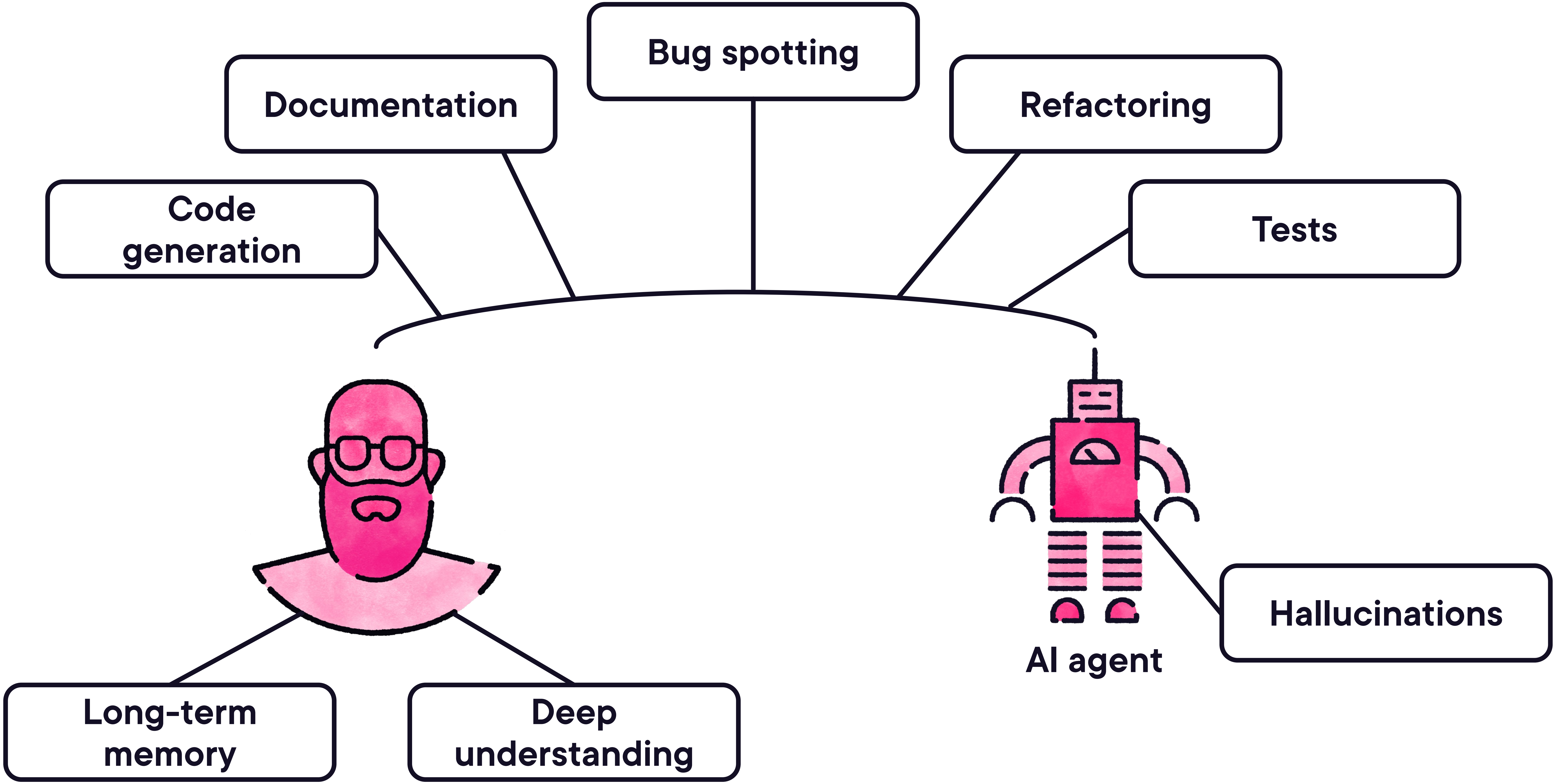
Type of AI Agents

Github Copilot

Jasper AI

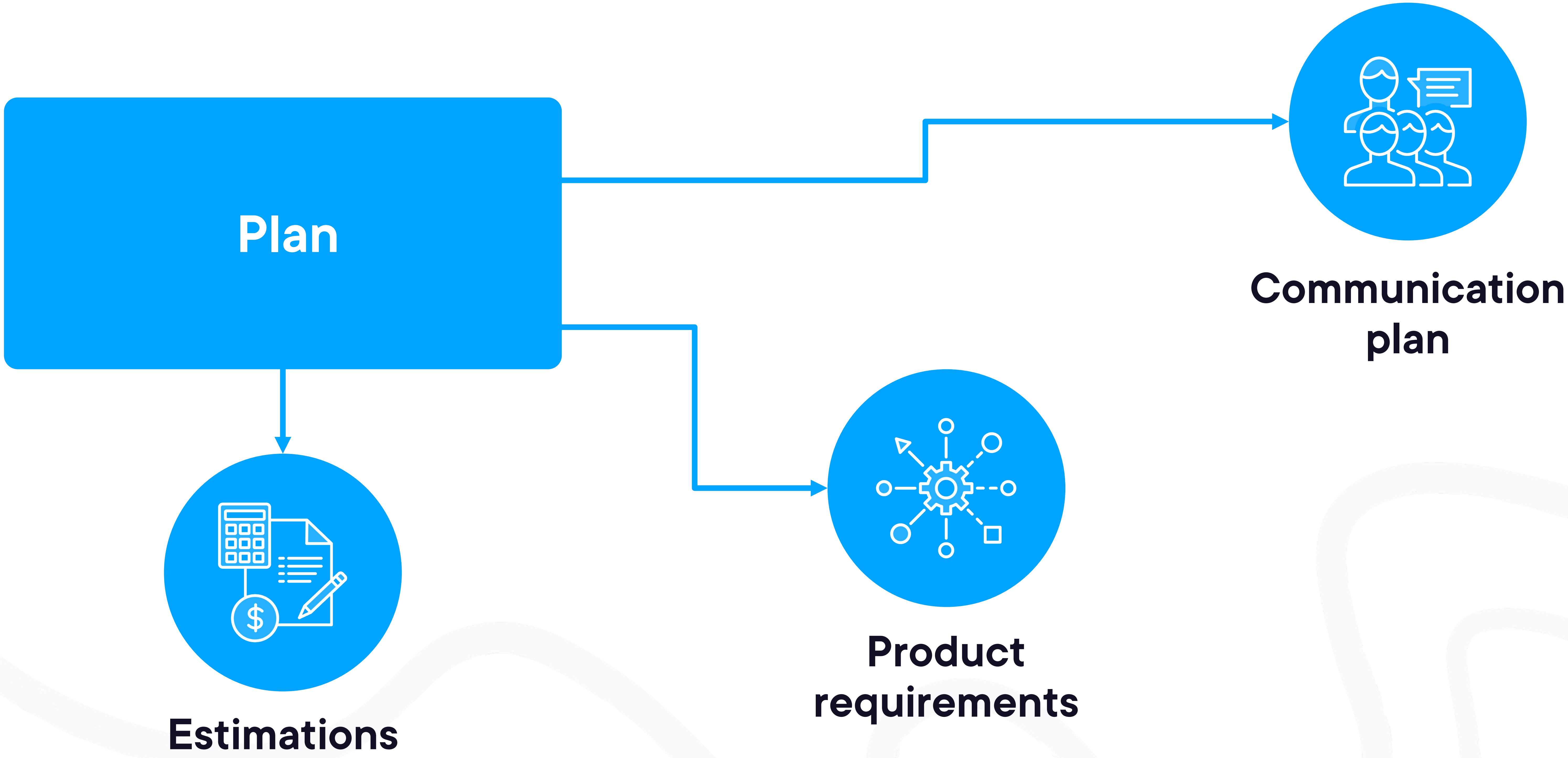
Copy.ai

Domain specific

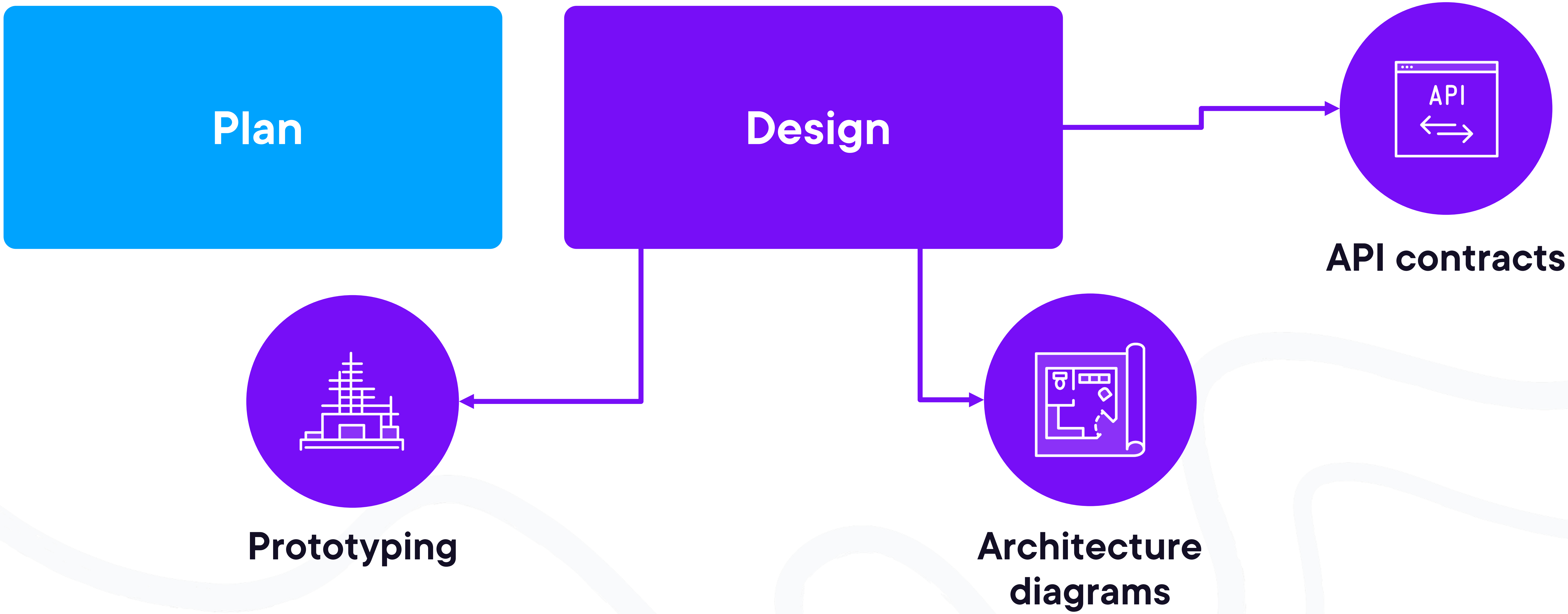

Code generation
Documentation
Bug spotting
Refactoring
Tests
Long-term memory
Deep understanding
AI agent
Hallucinations

AI in SDLC

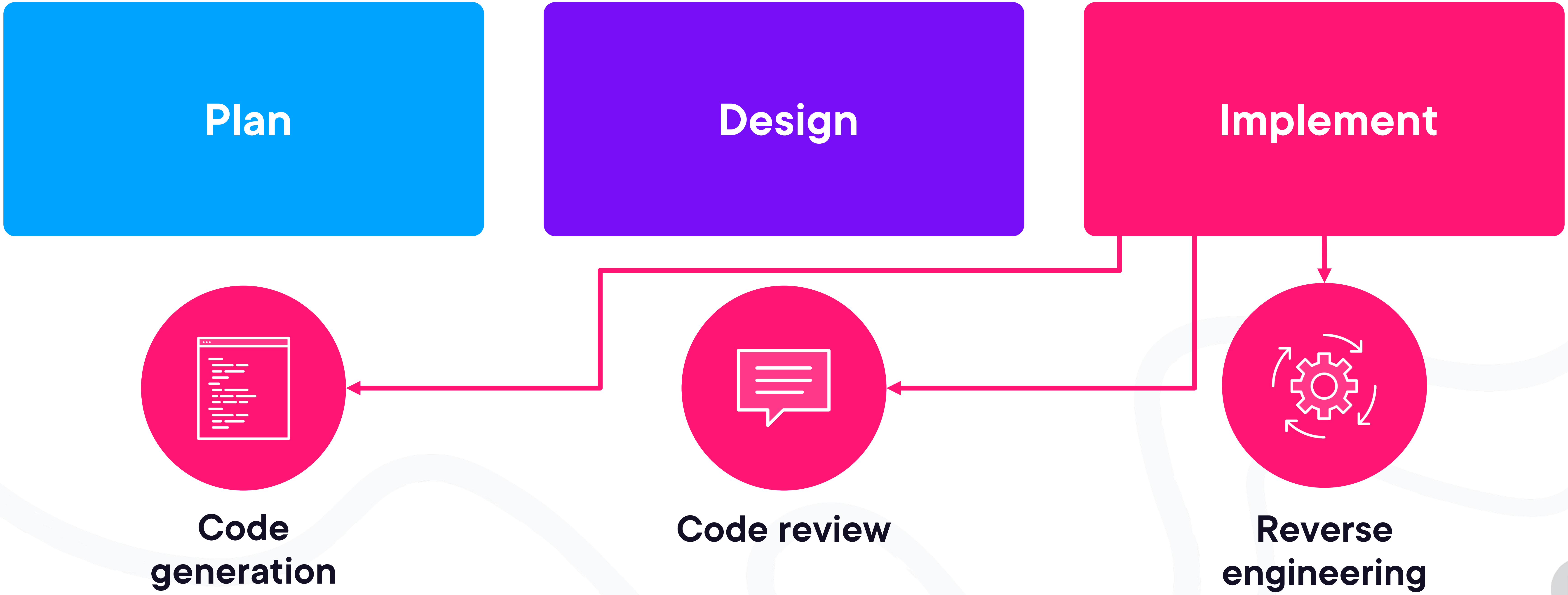
Software Development Lifecycle



Software Development Lifecycle



Software Development Lifecycle



Software Development Lifecycle

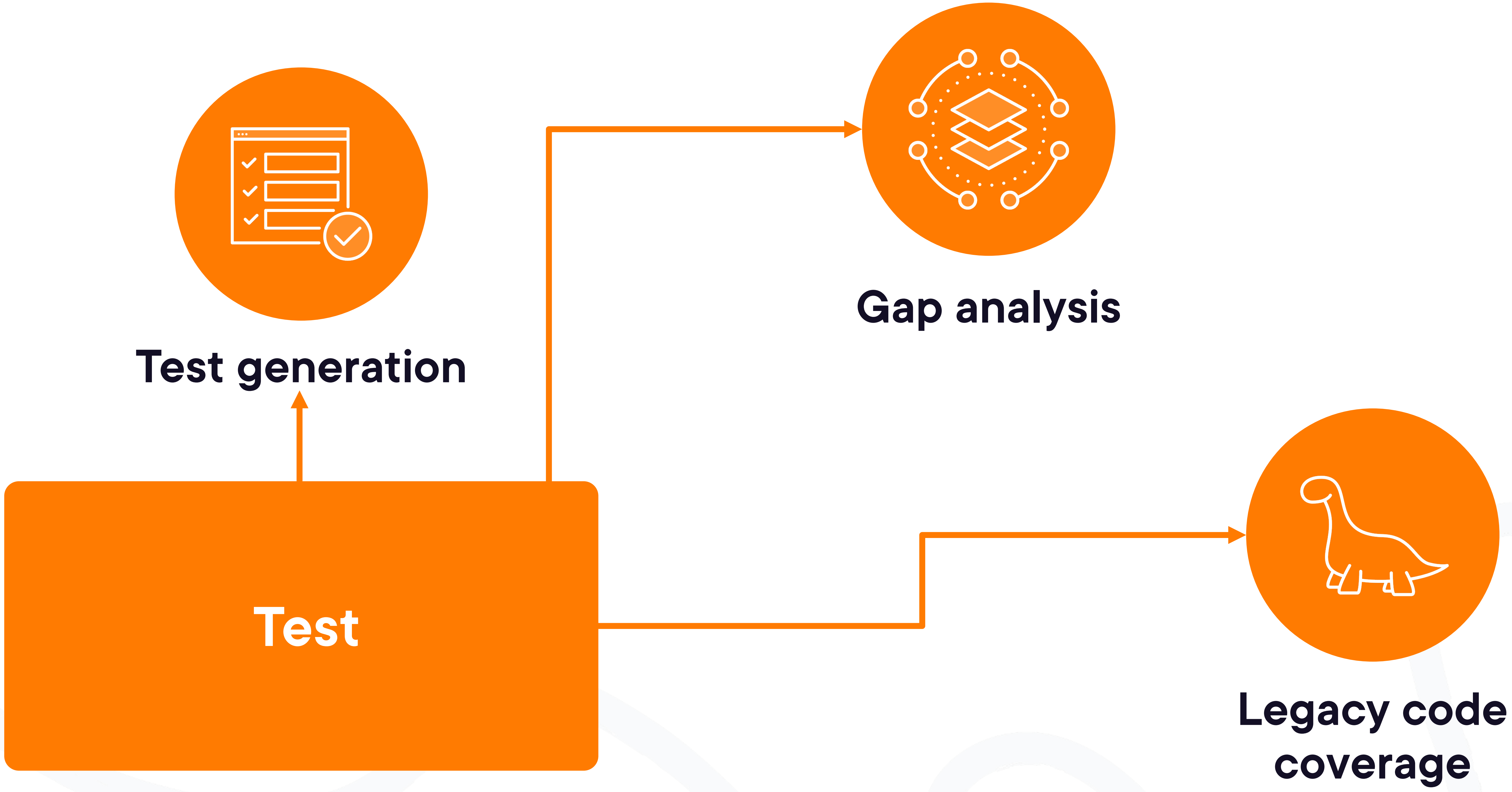
Plan

Design

Implement

Test

Software Development Lifecycle



Software Development Lifecycle

Plan

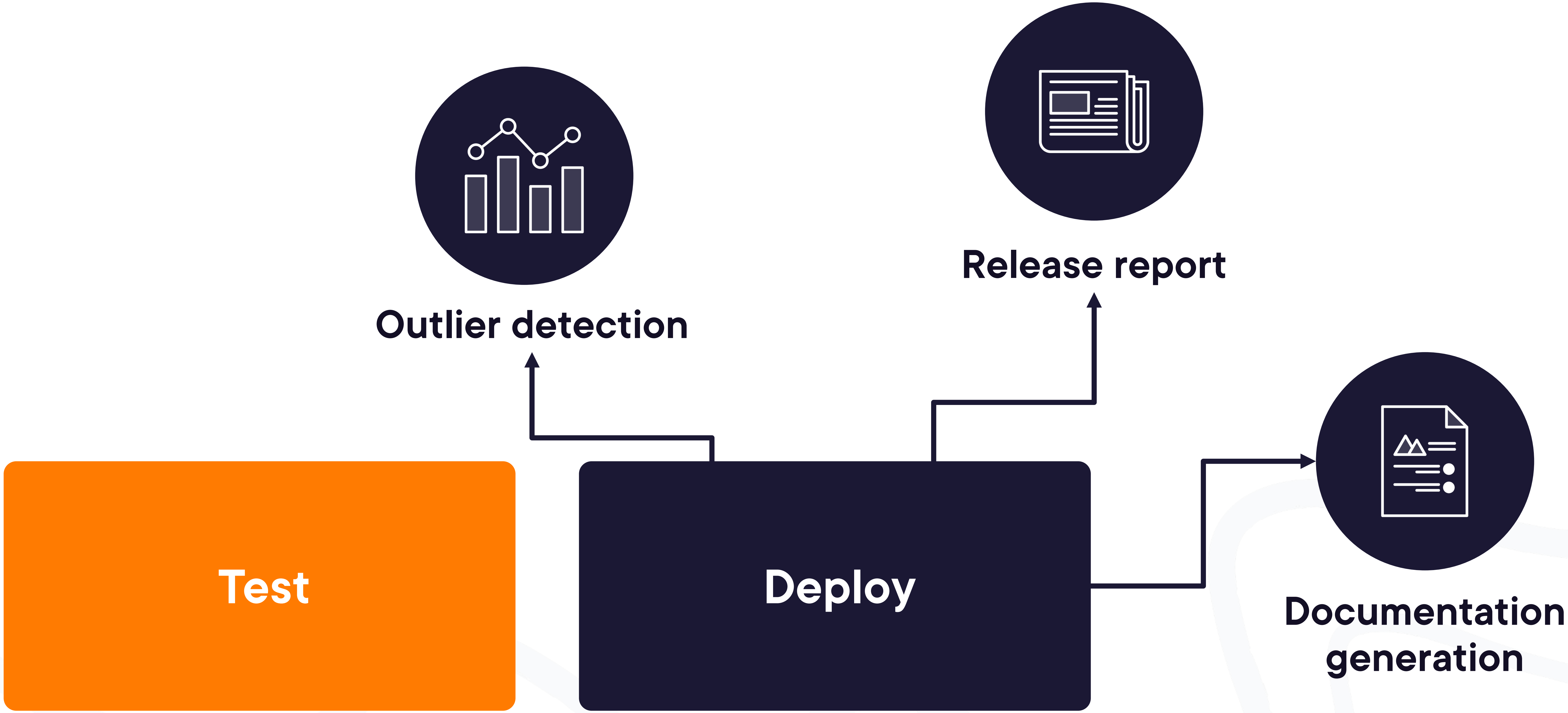
Design

Implement

Test

Deploy

Software Development Lifecycle



Software Development Lifecycle

Plan

Design

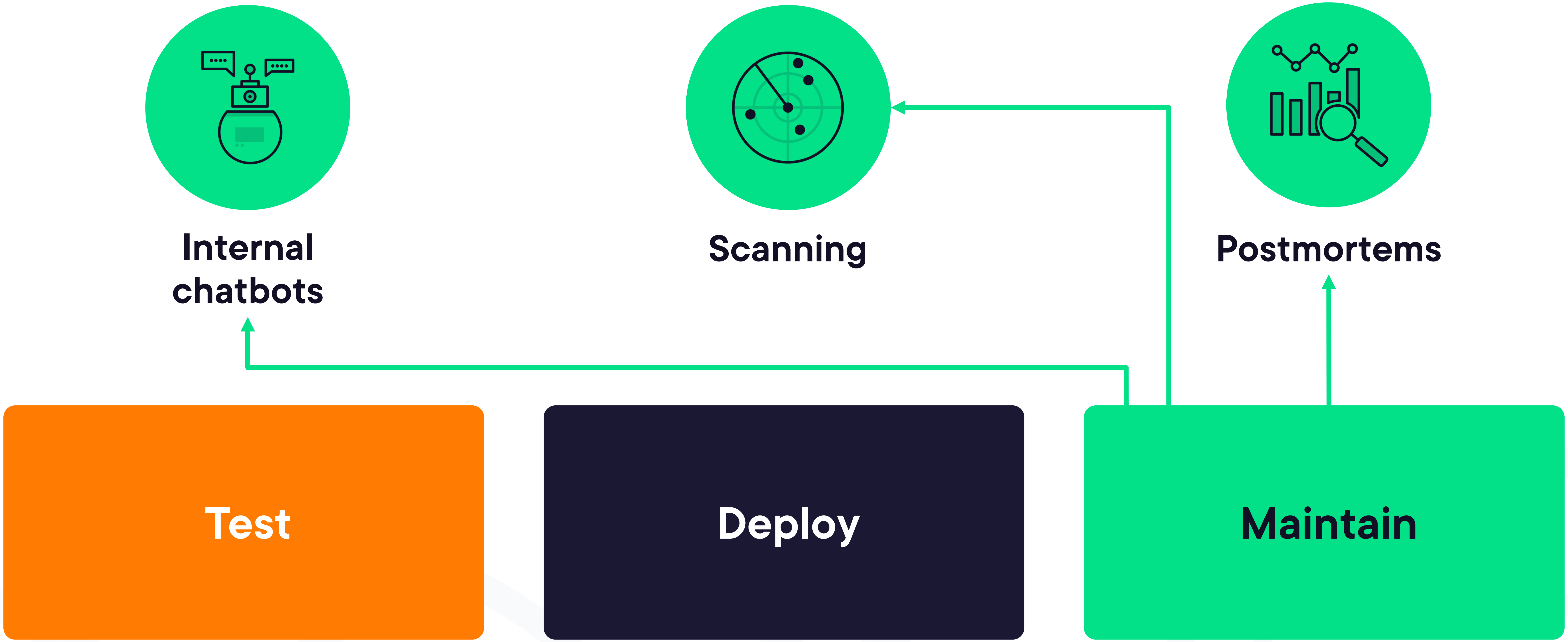
Implement

Test

Deploy

Maintain

Software Development Lifecycle



Engineering Productivity

Engineering Productivity

Metrics

Number of line of code

Time saved per task

Frequency of AI usage

Reduction in headcount

Numbers of features shipped

Engineering Productivity Frameworks

DORA

SPACE

DevEx

DORA Metrics

Change lead time

Deployment frequency

Change fail percentage

Mean time to recover

SPACE Metrics

Satisfaction

Performance

Activity

Communication and collaboration

Efficiency and flow

Developer Experience (DevEx) Metrics

Feedback loops

Cognitive load

Flow state

Pitfalls

Setting metrics as goal

Having one metric to rule them all

Using industry as a shield against improving

Making disparate comparisons

Having siloed ownership

Competing

Engineering Productivity Platforms

DX

Code Climate

Swarmia

LinearB

Haystack

Pluralsight Flow

Prompting for Precision

?

?

?

You

Write a Python function to validate an email.

Context

?

?

You

I’m building a backend for a user registration form.
Write a Python function to validate an email.

Context

Role

?

You

I’m building a backend for a user registration form.
You are a senior Python backend engineer.
Write a Python function to validate an email.

Context

Role

Expectation

You

I’m building a backend for a user registration form.
You are a senior Python backend engineer.
Write a robust, production-ready Python function that validates email addresses using regex. Include comments explaining each part of the function.

Context

Role

Expectation

You

The topic today is Big O notation.

Explain the concept in a way that's accessible to beginners with little to no math background. Use analogies and simple examples to help them intuitively grasp the idea.

Context

Role

Expectation

You

The topic today is Big O notation.
You're an expert in computer science education, teaching first-year university students.
Explain the concept in a way that's accessible to beginners with little to no math background. Use analogies and simple examples to help them intuitively grasp the idea.

Context

Reverse role

Expectation

You

The topic today is Big O notation.
You’re a university student who is just starting a course on algorithms.
Explain the concept in a way that's accessible to beginners with little to no math background. Use analogies and simple examples to help them intuitively grasp the idea.

Context

Reverse role

Expectation

You

You're confused about Big O notation and trying to make sense of it by asking questions and thinking aloud.
You’re a university student who is just starting a course on algorithms.
Express your confusion clearly, ask common beginner questions, and try to reason through the concept to understand it better.

Conflicting instructions

You

Explain this and summarize it so I can use it.

Lacks role

No format guidance

Too vague

Wrong level of detail: too shallow

Conflicting instructions

You

Lacks role

Explain this technical design document and summarize it so I can use it.

No format guidance

Too vague

Wrong level of detail: too shallow

Conflicting instructions

You

Act as a senior developer advocate.
Explain this technical design document and summarize it so I can use it.

Lacks role

No format guidance

Wrong level of detail: too shallow

Conflicting instructions

You

Act as a senior developer advocate.
I need help rewriting a technical design document into a summary that's easy to understand for junior developers.

No format guidance

Wrong level of detail: too shallow

You

We need to extend our internal documentation to make it more accessible to junior backend engineers.
Act as a senior developer advocate.
I need help rewriting a technical design document into a summary that's easy to understand for junior developers.

No format guidance

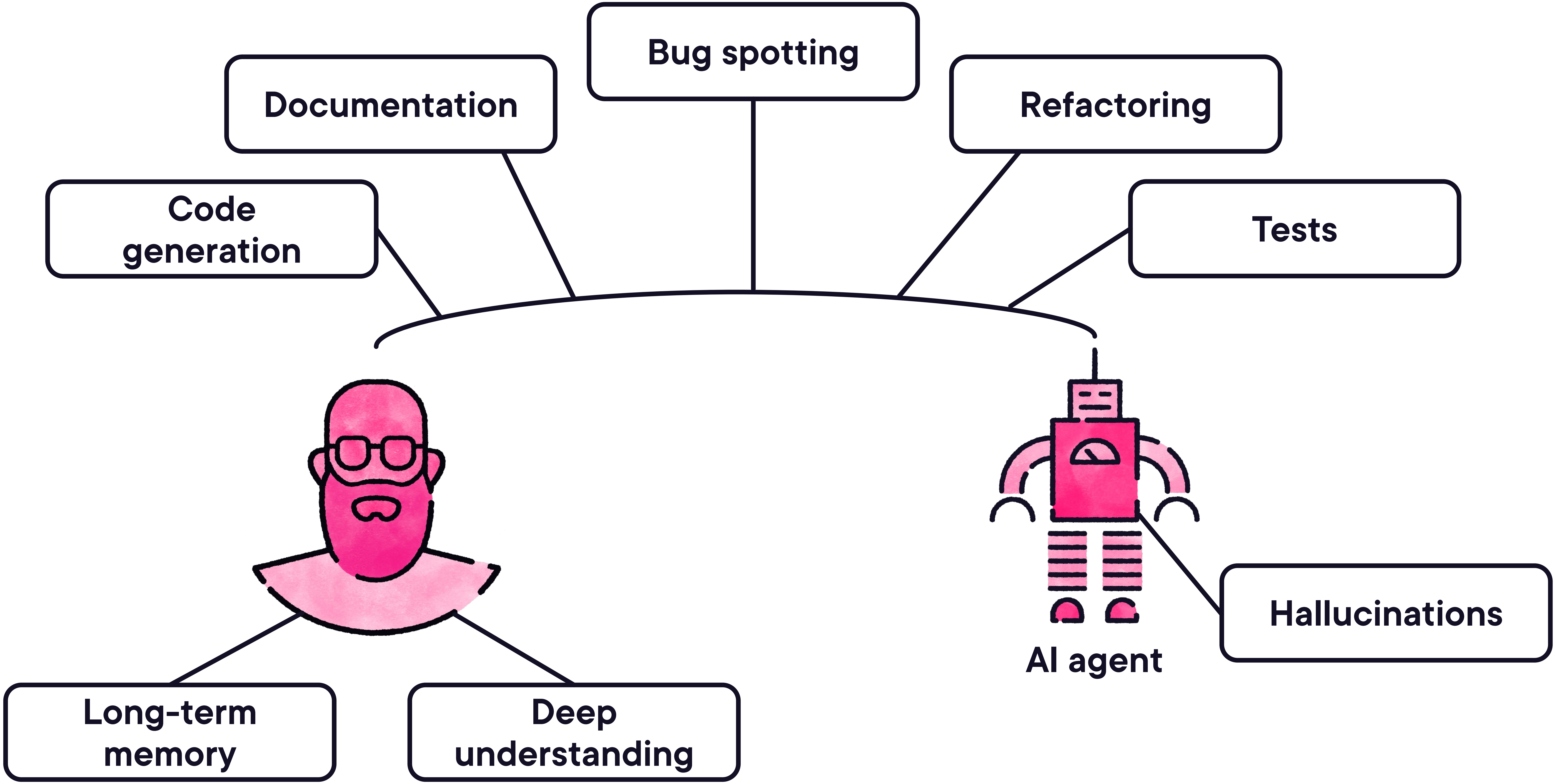
Wrong level of detail: too shallow

You

We need to extend our internal documentation to make it more accessible to junior backend engineers.
Act as a senior developer advocate.
I need help rewriting a technical design document into a summary that's easy to understand for junior developers.
Keep it concise, use plain language, and highlight the most important ideas. Include analogies or examples where useful.

No format guidance

Human-in-the-loop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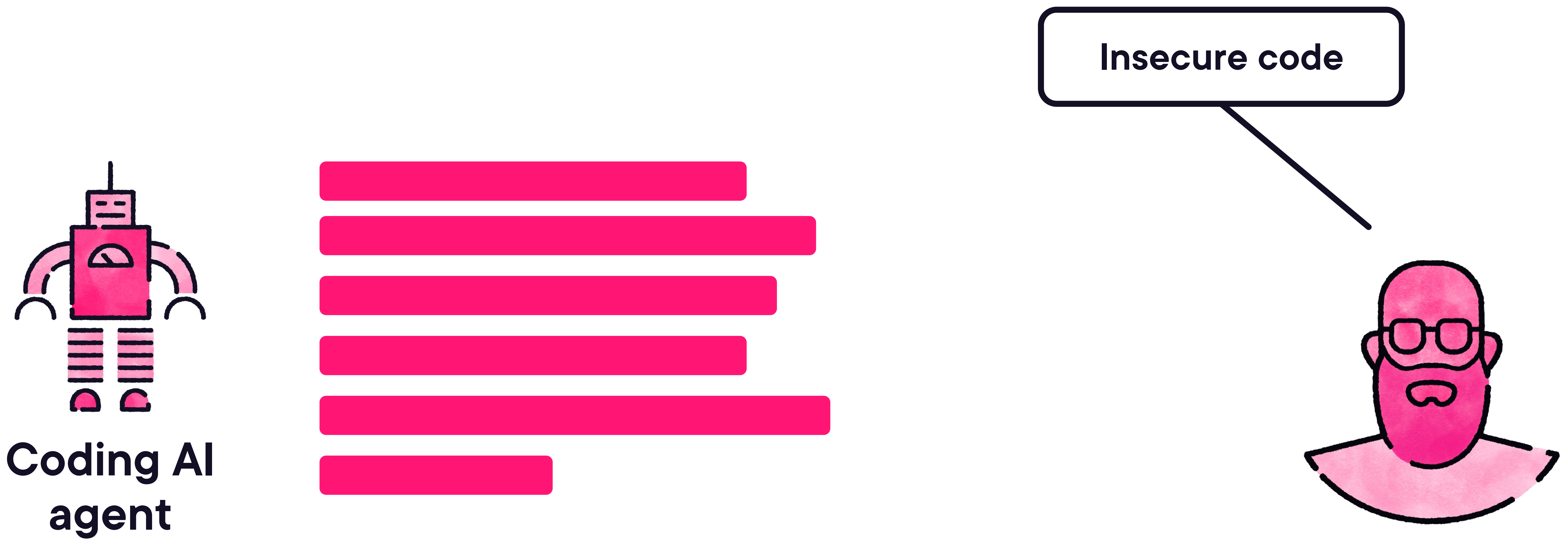
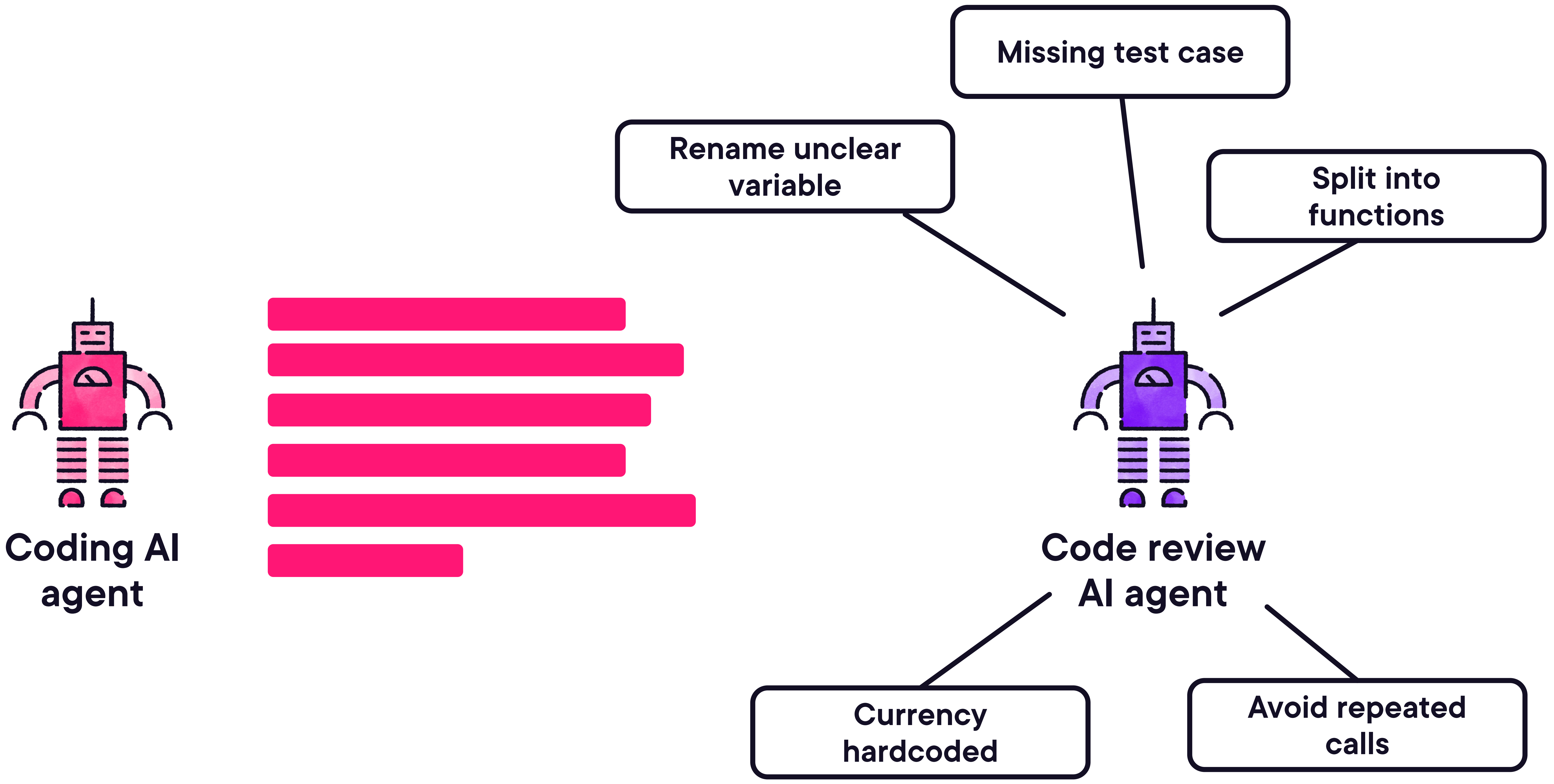
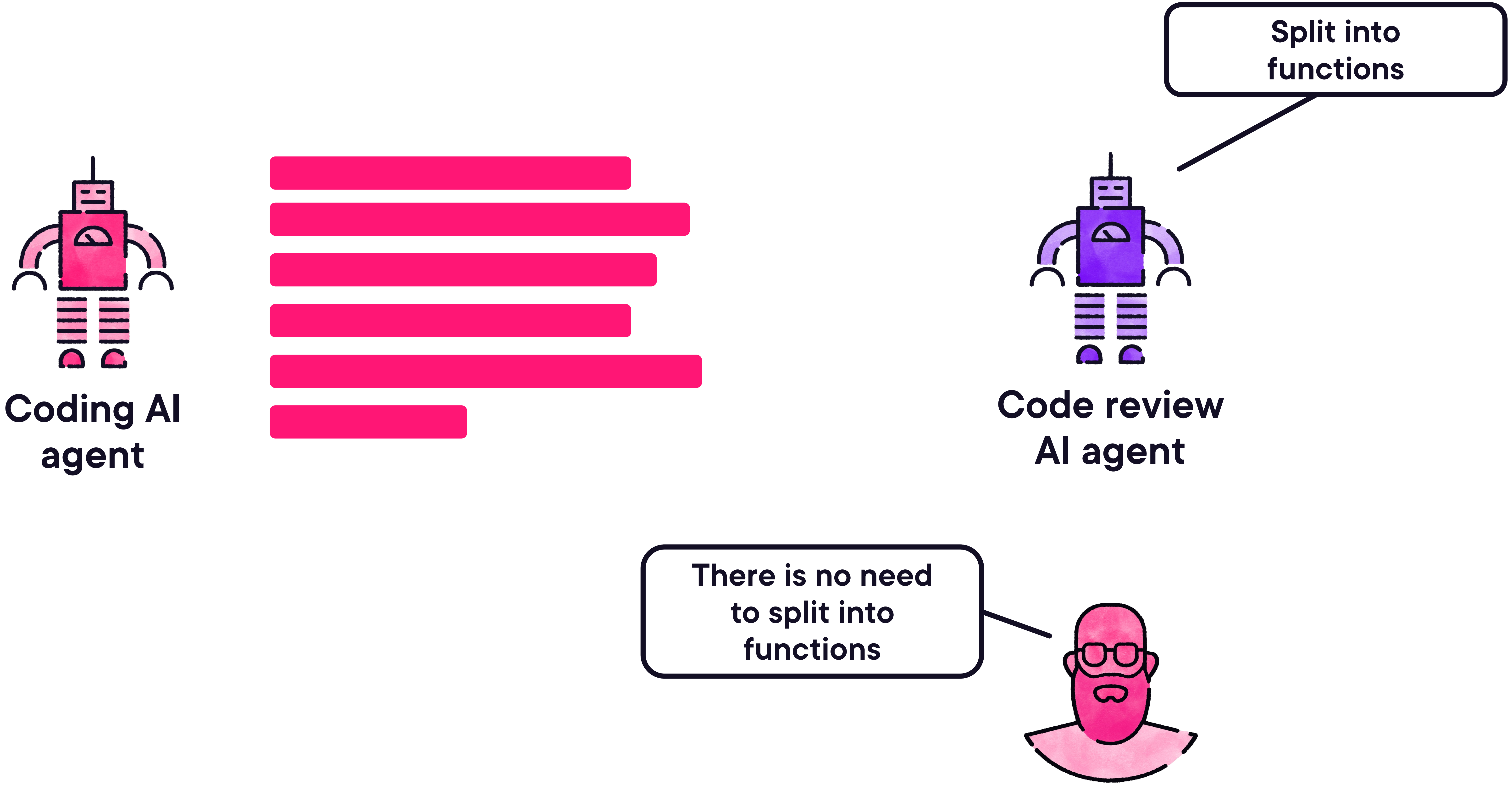
Hallucinations
Long-term memory
Deep understanding


Insecure code
Coding AI agent


Missing test case
Rename unclear variable
Split into functions
Coding AI agent
Code review AI agent
Currency hardcoded
Avoid repeated calls

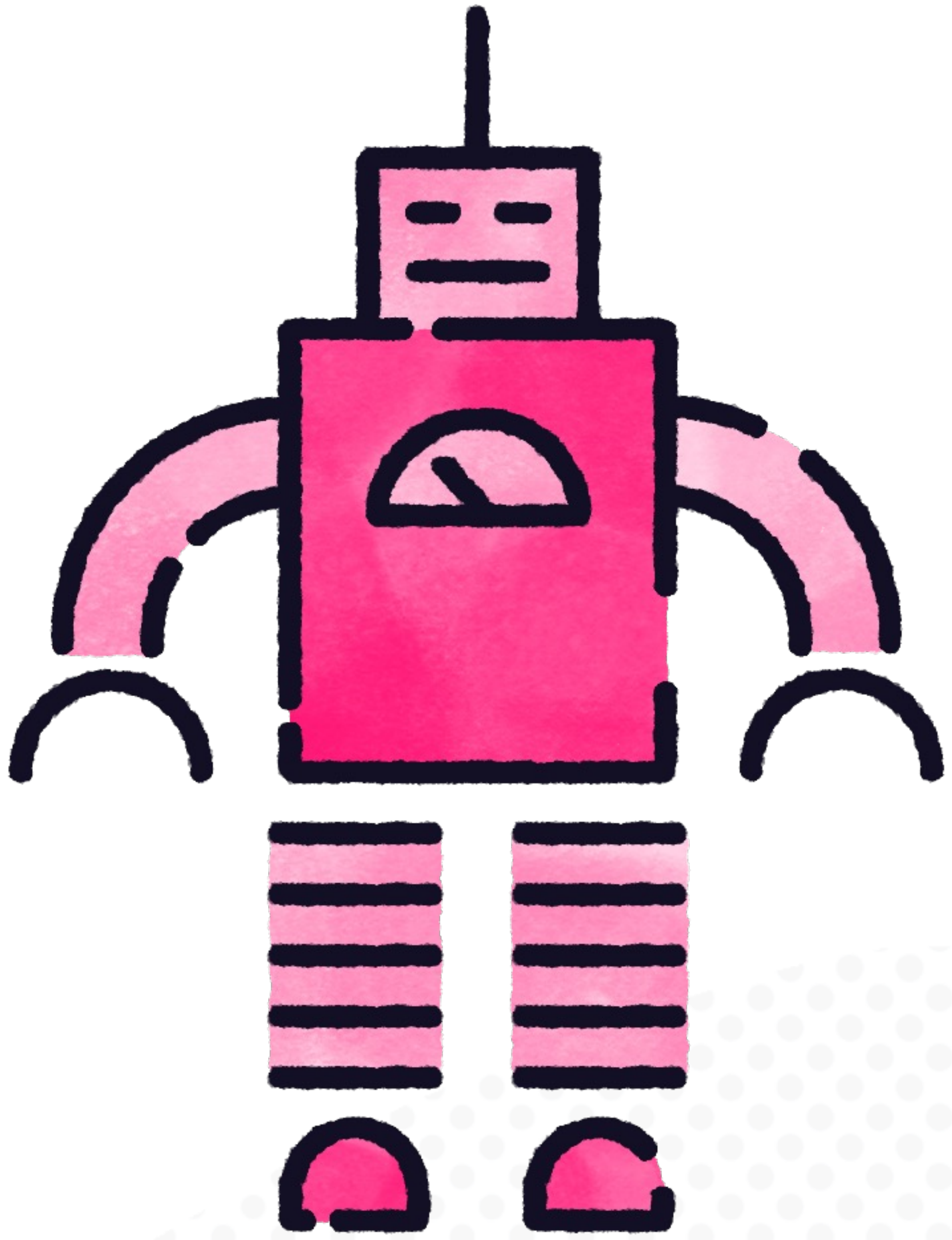


Split into functions
Coding AI agent
Code review AI agent
There is no need to split into functions

Junior colleague

Delegate repetitive tasks

Check when business knowledge is needed



Best Practices

Code readability

Code reviews

Testing

Formatting

Static code analysis

Demo: Technical Decision Justification

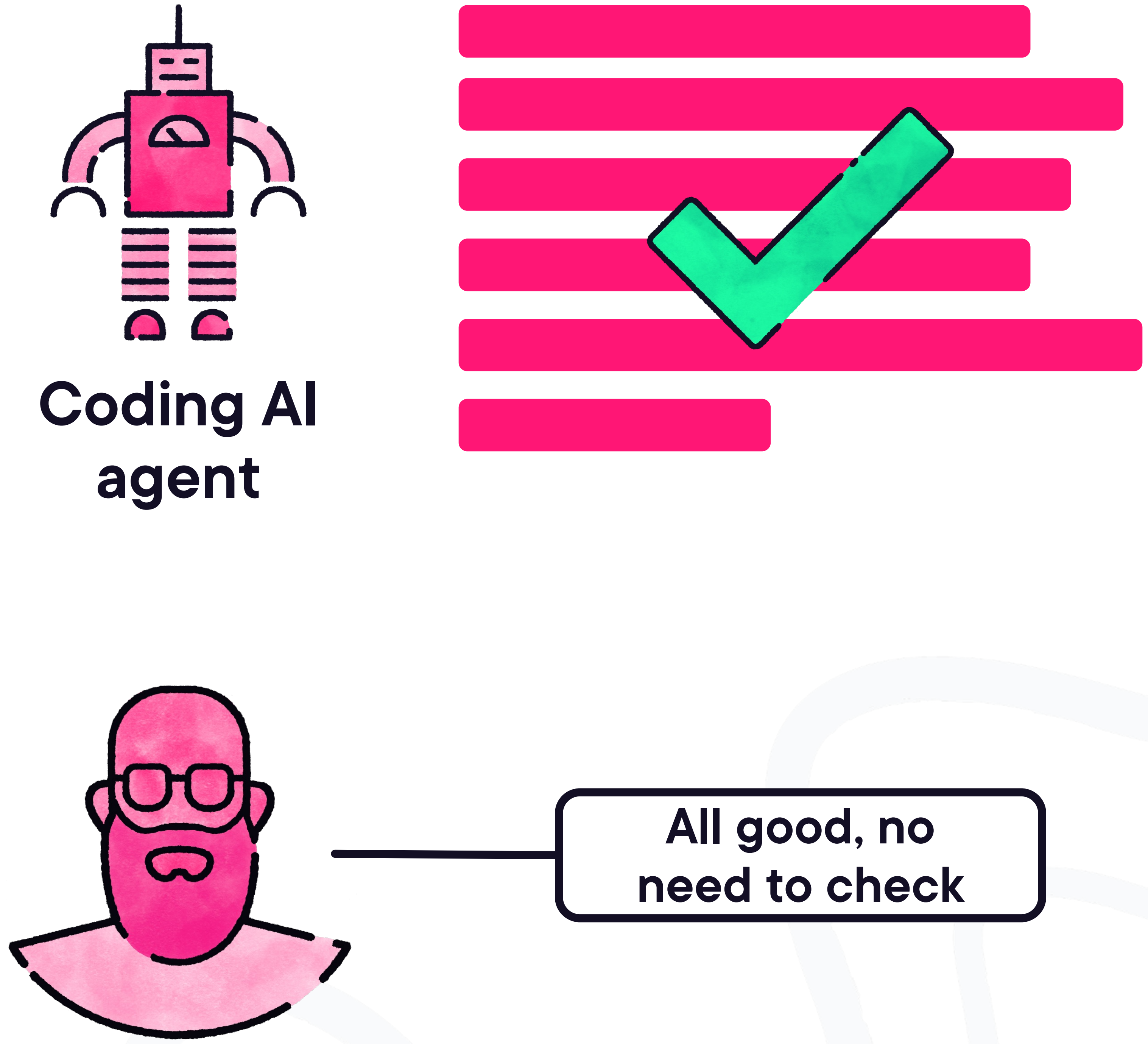
Demo: Implementation with AI Agents

Demo: Refining Development Tickets

Overuse and Underuse

Signs of Overuse

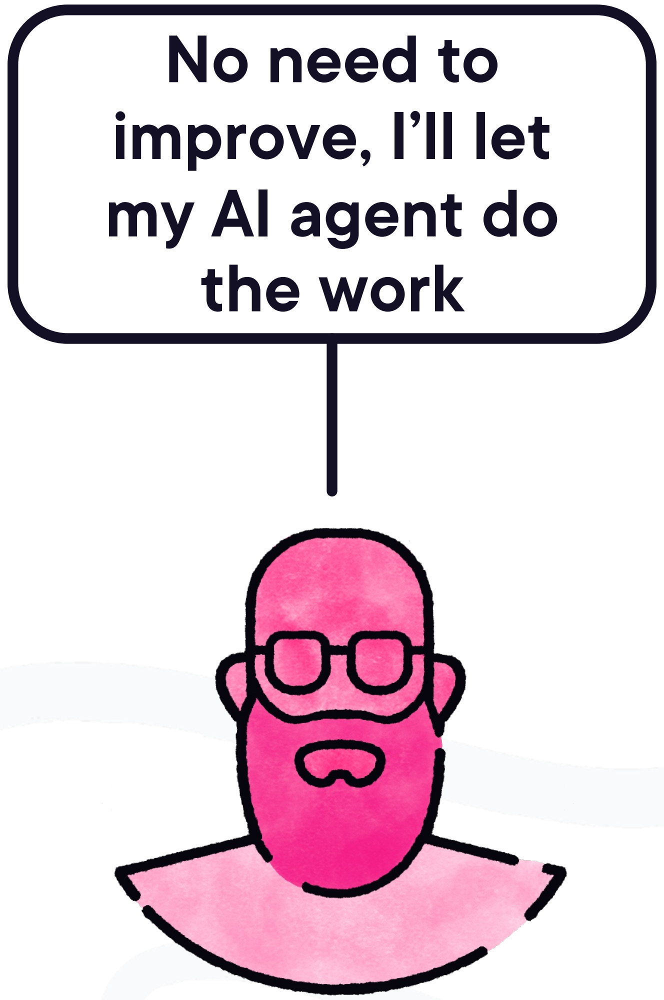
Automation bias



Signs of Overuse

Automation bias

Degraded skills



Signs of Overuse

Automation bias

Degraded skills

Shipping low quality products

DORA Metrics

Change lead time

Deployment frequency

Change fail percentage

Mean time to recover

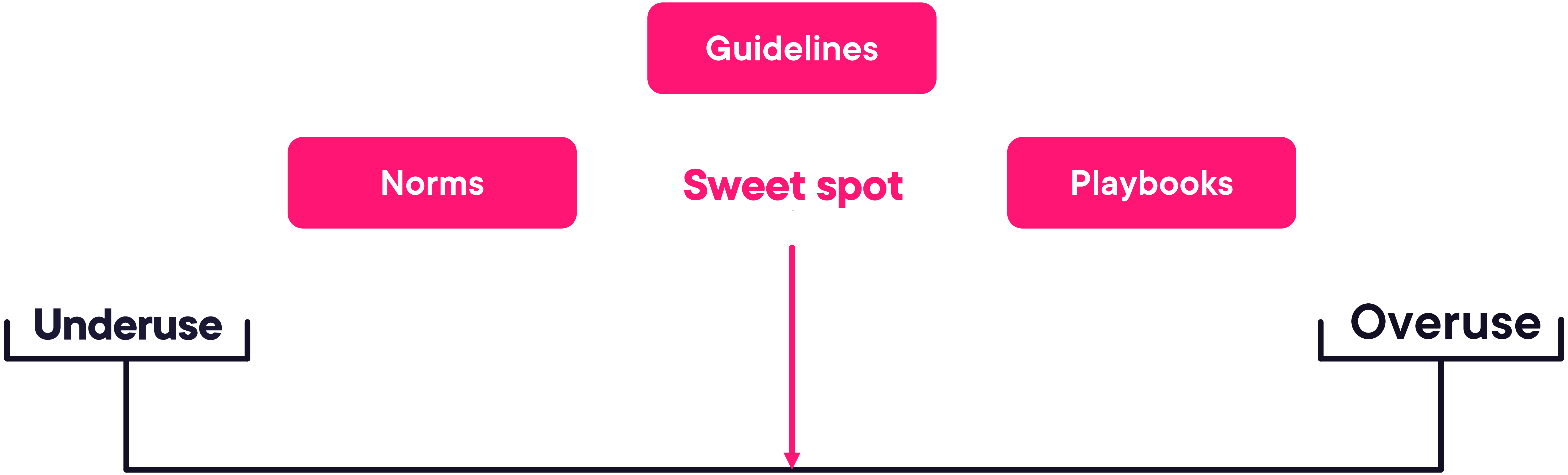
Signs of Underuse

Manual toil

Missed leverage

Context switch overload

AI Agents Usage



Thank you for your time!

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